Fine-mapped eQTL evaluation using XQTL mega-eQTL data

1 Overview

This analysis applies the variant-effect benchmark in `eqtl.tex` to six cell-type mega-eQTL fine-mapping datasets supplied as Sum of Single Effects (SuSiE) posterior inclusion probabilities (PIPs) for expression quantitative trait loci (eQTLs). In this report, XQTL names this six-group source. The groups are astrocytes (Ast), excitatory neurons (Exc), inhibitory neurons (Inh), microglia (Mic), oligodendrocyte precursor cells (OPC), and oligodendrocytes (Oli). The input does not contain an endothelial group. We compare these results with the existing singlebrain fine-mapping, using its broad Ast, Ext, IN, MG, OPC, and OD panels, respectively.

2 Methods

For both data sources, gene identifiers are version-normalized and matched to the same GENCODE hg38 gene annotation. Variants are restricted to single-base ACGT substitutions on the gene chromosome within 512 kb of the gene transcription start site (TSS). Variants with PIP > 0.75 are positives; those with PIP < 0.01 are negatives. The benchmark retains every positive and samples up to 1,000 negatives per group with a fixed seed. Each model receives a 1 Mb sequence window centered at the gene TSS.

The trained comparisons reuse the head-only, LoRA, LoRA+LoCon, and raw-Zemke LoRA+LoCon checkpoints and RNA track mappings in `eqtl.tex`. Native AlphaGenome mutation effects use all available RNA-seq and CAGE tracks, followed by the same biologically constrained track selection used in that report. Performance includes area under the receiver operating characteristic curve (AUROC) and average precision (AP), overall and by distance to the TSS; random AP equals the positive prevalence within each distance bin. CAGE means Cap Analysis of Gene Expression.

Fine-mapping agreement is computed on gene-variant keys consisting of gene, chromosome, one-based position, reference allele, and alternate allele. Exact high-confidence overlap uses all eligible variants and reports intersection, Jaccard similarity, and directional recall at PIP > 0.75.

3 Fine-mapping agreement

The full scan covered 70.9–218.7 million input rows per group. After matching to an annotated gene, requiring a single-base substitution on the gene chromosome, and applying the TSS window, 14.3–41.2 million records per group remained. Table 1 gives counts and PIP classes.

At the high-confidence threshold, exact gene-variant Jaccard similarity is 0.080–0.118 (Table 2). XQTL positive recall against singlebrain is 0.28–0.37, while singlebrain positive recall against XQTL is 0.09–0.16. Thus the sources share some high-confidence calls but the two sets are not interchangeable.

Table 1: XQTL source counts. Eligible rows satisfy the common gene, allele, chromosome, and TSS-window filters.

Group	Input rows	Eligible rows	PIP > 0.75	PIP < 0.01
Astrocyte	167,399,779	31,850,543	581	31,769,295
Excitatory	218,708,555	41,150,681	1,784	40,976,915
Inhibitory	207,868,448	39,394,880	813	39,281,800
Microglia	70,865,864	14,332,701	168	14,305,646
OPC	122,683,778	24,317,596	358	24,269,005
Oligodendrocyte	178,552,659	33,830,282	852	33,738,029

Table 2: Exact high-PIP overlap. Recall columns use the named source as denominator.

Group	XQTL n	Singlebrain n	Shared	Jaccard	XQTL recall	SB recall
Astrocyte	581	1,231	172	0.105	0.296	0.140
Excitatory	1,784	3,420	504	0.107	0.283	0.147
Inhibitory	813	1,838	236	0.098	0.290	0.128
Microglia	168	668	62	0.080	0.369	0.093
OPC	358	929	122	0.105	0.341	0.131
Oligodendrocyte	852	1,820	283	0.118	0.332	0.155

4 Variant-effect performance

The completed benchmark scores 10,452 of the 10,556 cached records. The remaining 104 records are variants that could not be represented in the reference sequence and are excluded consistently from model and native-track comparisons. These results use the corrected Zemke 2023 RNA track lookup described in `eqtl.tex`; the earlier version scored microglia, oligodendrocyte, OPC, and inhibitory panels with other cell types' tracks.

Table 3 reports AUROC and AP pooled over panels and AUROC averaged over the six panel groups after computing it within each group. The native series uses one biologically constrained native RNA-seq or CAGE track selection per XQTL panel, chosen on these variants. The raw-Zemke LoRA+LoCon model has the highest pooled TSS-bin and gene-span AUROC and the highest TSS-bin AP. The native matched tracks have the highest within-group AUROC and gene-span AP, subject to selection optimism. Trained models differ by at most 0.033 in within-group TSS AUROC and 0.003 in within-group gene-span AUROC. Distance-stratified performance is shown in Figures 4.1 and 4.1.

Table 3: Aggregate XQTL variant-effect performance across the six panels. "Within" is the mean of per-group AUROC.

	TSS bin			Gene span		
Method	AUROC	AUPR	Within	AUROC	AUPR	Within
Head-only	0.536	0.498	0.538	0.635	0.608	0.644
LoRA	0.534	0.503	0.555	0.628	0.600	0.642
LoRA+LoCon	0.533	0.496	0.543	0.637	0.615	0.643
LoRA+LoCon, raw Zemke ATAC	0.558	0.545	0.571	0.643	0.623	0.646
Native matched tracks	0.541	0.491	0.577	0.627	0.626	0.650

4.1 Trained track specificity

The specificity statistic D in `eqtl.tex` is the mean AUROC of each panel group scored with its own track-group composite minus the mean AUROC with other groups' composites. With six groups

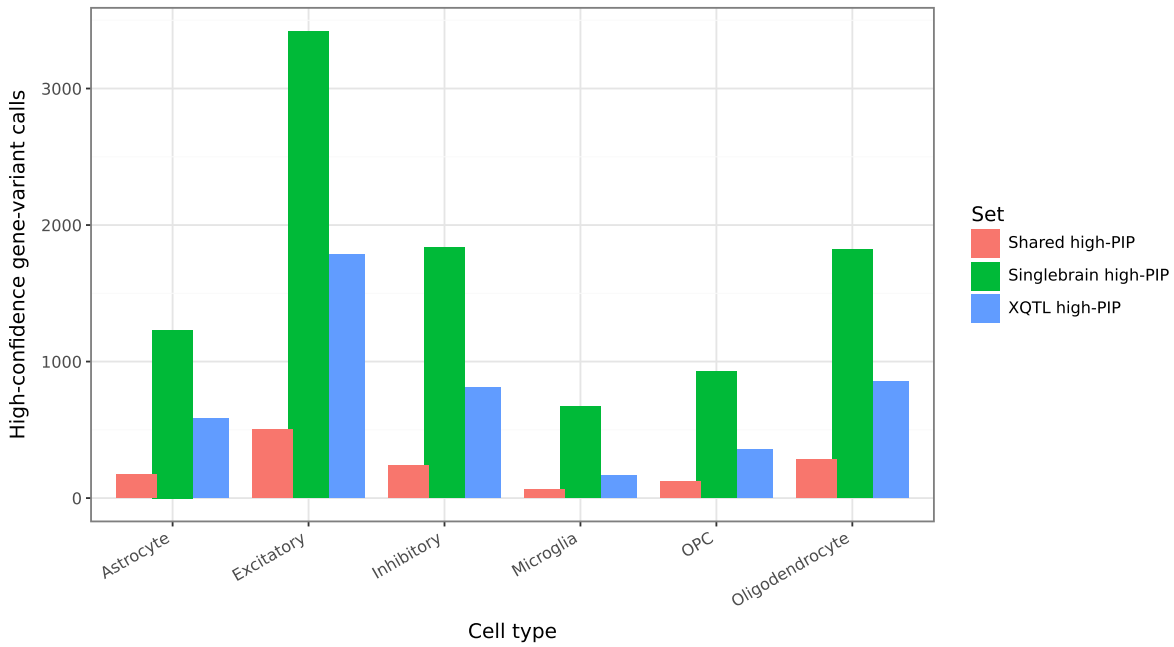


Figure 1: Counts of eligible high-PIP gene-variant calls in each source and their exact intersection.

the exact permutation p-value has a minimum of $1/720 \approx 0.0014$. D is positive for every model and summary (Table 4). On the gene-span summary it is 0.033 to 0.057 and reaches the minimum p-value for all models; on the TSS bin it is 0.020 to 0.030 with $p \leq 0.028$. Microglia panels are best discriminated by the microglia composite in every model and summary, with gene-span AUROC 0.69 to 0.73 compared with 0.54 to 0.63 for other composites (Figure 2). Astrocyte and oligodendrocyte panels rank their own composite first in six and five of eight combinations. Excitatory panels rank their own composite first in four combinations, but never by more than 0.006 in AUROC. OPC panels rank it first on the gene-span summary for three models and fourth or fifth on the TSS bin, and inhibitory panels never rank it first for any model on the gene span.

Table 4: Cell-type specificity of trained RNA tracks on XQTL panels. D is the matched minus unmatched mean AUROC and p its exact permutation p-value; ρ is the median Spearman correlation between track effects.

	TSS bin			Gene span		
Model	D	p	ρ	D	p	ρ
Head-only	0.020	0.0278	0.70	0.033	0.0014	0.69
LoRA	0.028	0.0097	0.72	0.056	0.0014	0.68
LoRA+LoCon	0.021	0.0125	0.71	0.057	0.0014	0.69
LoRA+LoCon, raw Zemke ATAC	0.030	0.0069	0.72	0.052	0.0014	0.69

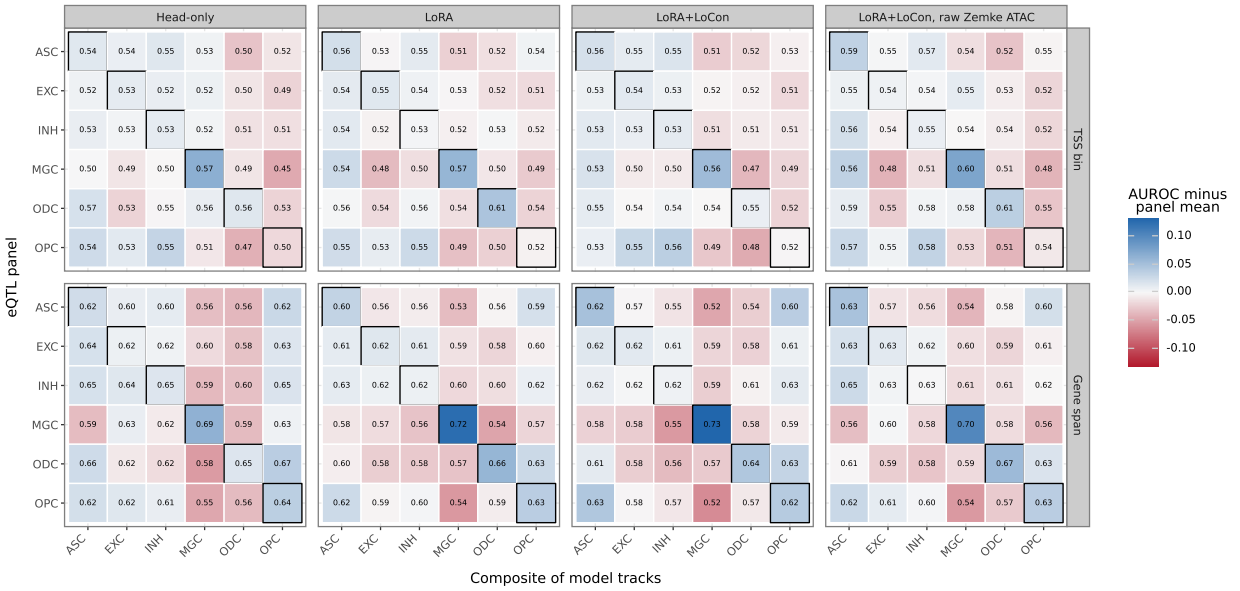


Figure 2: AUROC of each XQTL panel group (rows) scored with each track-group composite (columns). Color shows AUROC minus the row mean; black outlines mark the matched composite.

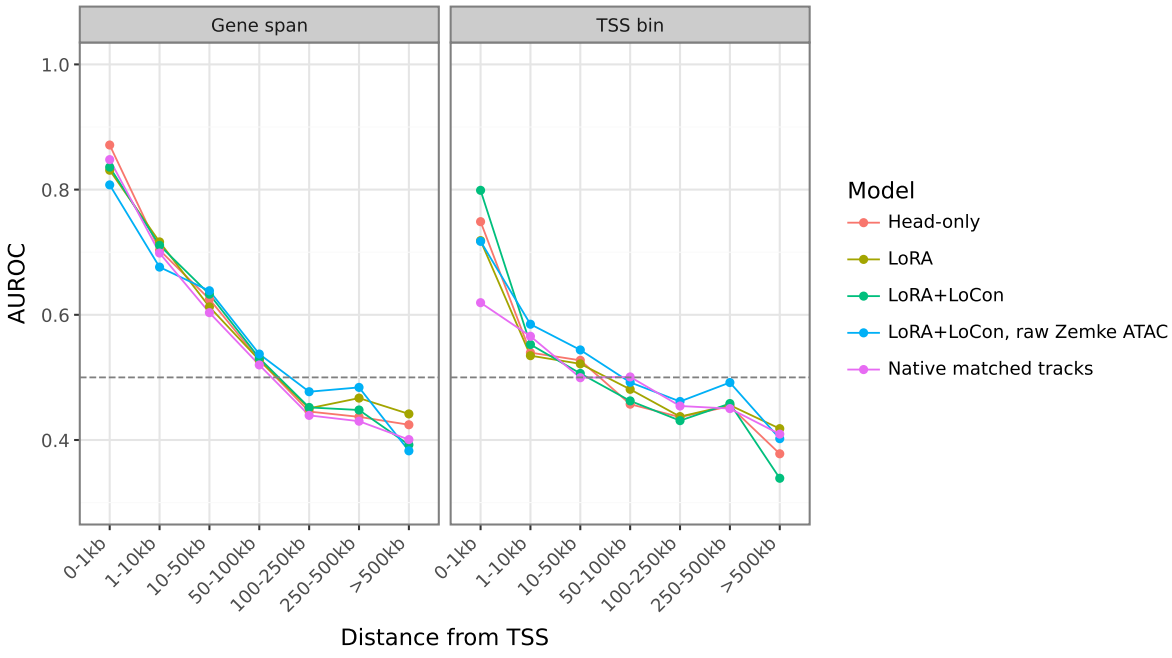


Figure 3: AUROC by absolute distance from the TSS.

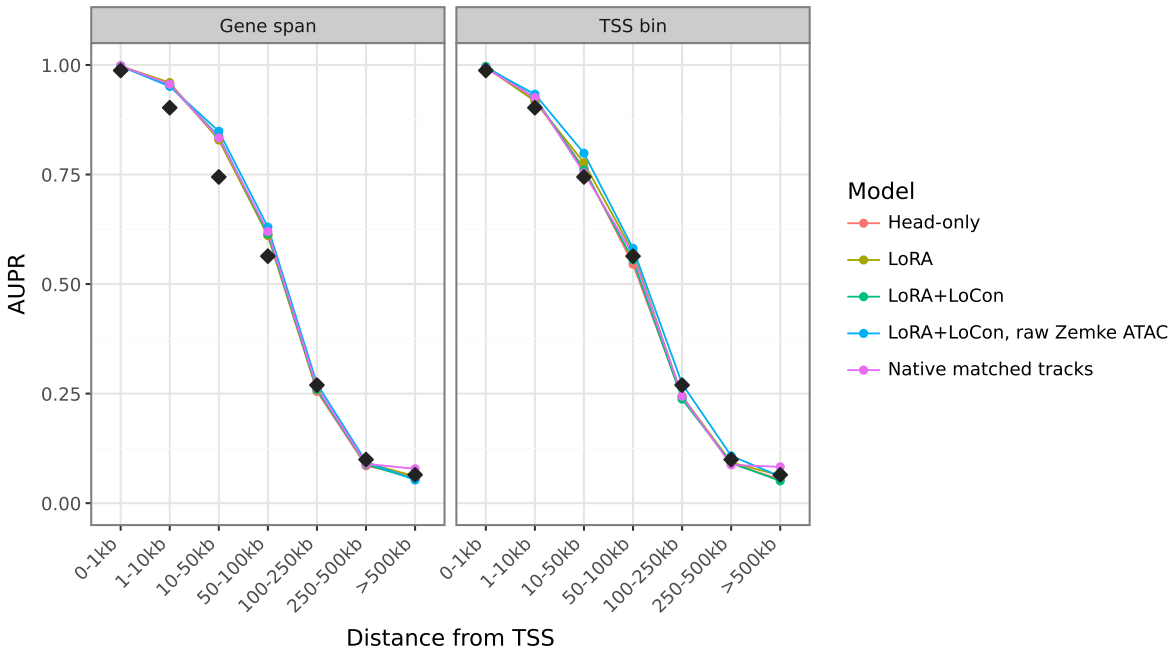


Figure 4: Average precision by absolute distance from the TSS. The random reference is the positive prevalence in each bin.